AI61004 Statistical Foundations of AI/ML Quiz 1 (Set 1)

Date 12th February 2026 **Time 15:00 AM -15:40 PM** **Full Marks: 20**

Name: ____ Roll: ____________________

Q1. A random variable X is sampled from a Bernoulli distribution Ber(p). If X=1, then Y is sampled from a Gaussian distribution N(u, s), else Y is sampled from a Gamma distribution Γ(a, b). Calculate the mean and variance of Y. **[2+5=7 marks]**

Solution sketch: We can write Y = X*Y1 + (1-X)*Y2, where Y1~N(u, s) and Y2~ Γ(a, b). Clearly, X, Y1, Y2 are independent of each other. E(Y) = E(X)*E(Y1) + (1-E(X))*E(Y2) = pu + (1- p)*(a/b), as E(Y2) = a/b.

Var(Y) = $E(Y^2) - E(Y)^2$. So we need to derive expression of Y^2, which is $X^2Y1^2 + (1-X)^2Y2^2 + 2*X*(1-X)*Y1*Y2$. Its expectation can be calculated again by applying linearity of expectations. $E(X^2) = p$, $E(Y1^2) = u^2 + s^2$, $E(Y2^2) = (a+a^2)/b^2$

Q2. Calculate the Moment Generating Function of Uniform Distribution U(1,5). Using it, find the variance. Repeat the same for Poisson(5). **[5+5=10 marks]**

Solution sketch: For X~U(1,5), the PDF is ¼ in the interval (1,5). E(exp(tX)) is obtained by integrating exp(tX)*(1/4) in the limits (1,5), which gives (exp(5t)-exp(t))(4t).

In case of Poisson, it is not elegant. Since it is a DRV whose support is (0,1,2….), the MGF will be an infinite series. By taking the exp(-5) term as common, this infinite series becomes an exponential series of the form $(5e^t)^x/x!$ which sums to $\exp(5e^t)$. Finally, the MGF comes to exp(5*exp(t)-5).

Expanding exp(tX), MGF equals $1 + tE(X) + t^2E(X^2)/2! + \dots$. Calculating first derivative and setting t=0 gives E(X), and calculating second derivative and setting t=0 gives $E(X^2)$. From these two, we get Var(X)

Q3. Calculate an expression for the K-L divergence between Beta(a,b) and U(0,1). **[3 marks]**

Solution sketch: Since the PDF of U(0,1) is just 1, it is just the entropy of Beta(a,b). It is not needed to calculate the closed form expression of that.
